



Armada Kalamunda Group

Intubation and Management of the Ventilated Patient in the Intensive Care Unit Guideline

1. Purpose

Nursing care of the ventilated patient will only be provided by Ventilator Competent Registered nurses.

The purpose of this guideline is to provide clinical staff with a working framework for the intubation and management of the ventilated patient.

2. Applicability

This guideline applies to medical and nursing staff working in the Intensive Care Unit at the Armada Kalamunda Group.

3. Intubation

The decision to offer Intubation in the critical care setting is the decision of a critical care trained medical professional. It is expected that they will apply their knowledge to determine the risks and benefits for an individual patient and communicate that to the patient or their carers to allow joint decision making, that is in the patient's best interests and aligns with the patient's goals of care and desired functional outcome.

Use of an endotracheal tube (ETT)

- enables the establishment and maintenance of a patent airway
- protects the patient from aspiration of gastric contents
- allows for controlled ventilation.

In the critical care setting intubation can be a planned or emergency procedure to facilitate mechanical ventilatory assistance.

Intubation is performed by trained medical staff or by staff undergoing training in intubation supervised by trained medical staff.

- All intubated/ventilated patients require a minimum nurse/patient ratio of 1:1
- All intubated/ventilated patients should have continuous cardiac, EtCO₂ and SpO₂ monitoring.
- Record observations on the ICU flow chart (hourly documentation minimum).
- Record relevant changes such as ETT status changes, position change, extubation or complications in the patient's integrated notes.

Intubation is a collaborative procedure which requires an experienced clinician in intubation as the team lead.

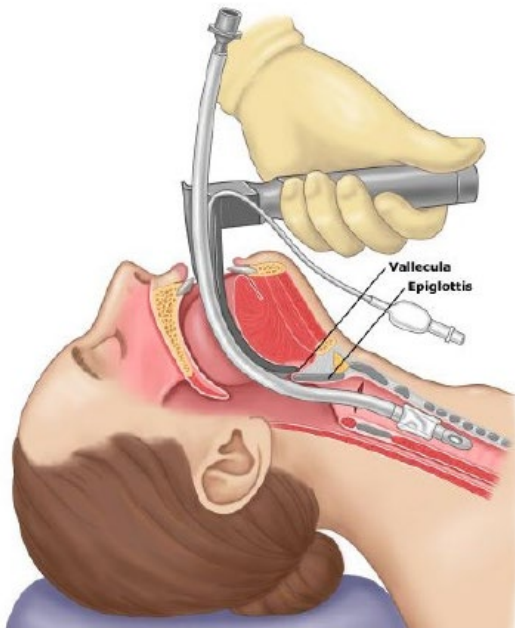


Figure 1 ⁶

3.1 The Endotracheal Tube

The size of ETT will vary depending on the patient's physical anatomy, airway view and clinical condition. ETTs vary in size from 6 – 8mm. 7.5 is suggested for females and 8.0 for males. The team leader will confirm the tube size required. AKG ICU is using ETTs capable of Suction above the cuff 7, 7.5 and 8 in size. Alternate sizes and ETT Types are available on the difficult airway trolley in ICU.

3.2 Equipment

- PPE as per standard and/or transmission-based precautions
- Standard ICU Airway (Blue) trolley and resuscitation (Red) trolley. Check that CICO kit is available on resuscitation trolley.
- Appropriately sized ETT and/or LMA if available. Laryngoscope with appropriate blade.
- Video laryngoscope
- Yankauer[®] suction catheter – operational and adjacent to patient
- Sterile water-based lubricant
- Introducer /Bougie/Stylet
- 10mL syringe
- ETT securing device - initially pre-cut tapes or pre-cut ribbon
- A self-inflating bag-valve mask (BVM) connected to oxygen @ 15L/min
- Modified Mapleson's C circuit (green bag)
- ETT catheter mount (i.e. Cobbs connector or Bodai)
- EtCO₂ circuit filter line connected to bedside monitor
- A ventilator attached to wall oxygen and air, and plugged into mains power, switched on and set to the prescribed settings PRIOR to intubation.

- Cuff manometer
- Stethoscope

3.3 Drugs for Intubation:

Potential standard medications drawn up and ready when decision to intubate is made:

- Propofol: 20mls 1% x 2
- Fentanyl 200 micrograms in 4mls
- Rocuronium 100mg in 10mls.
- Suxamethonium 200mg in 5mls
- Metaraminol 10mg diluted to 20mls = 0.5 mg/ml
- Midazolam 5mg
- Ketamine 200mg neat in 4mls

3.4 Drugs for infusion:

The patient's clinical condition often determines which medications are required for continuous infusion. Typically, Propofol is always used for continued sedation:

- 100mls Propofol 1%

Other analgesia often used can include:

- 500mcg Fentanyl in 50ml 10 mcg/ml
- 50mg of Midazolam in 50ml 1mg/ml
- 50mg Morphine 1mg/ml
- 200mg Ketamine 1mg/ml
- 200mcg Dexmedetomidine 4mcg/ml

3.5 Fluids

- 1L of Plasmalyte on a blood giving (pump) set attached to free-flowing intravenous cannula.

4. Procedure

4.1 Pre-preparation

Reassurance and explanation are given to patient/relatives/carer by a MO or Nurse.

- Ensure all other monitoring devices are attached and functional, including ECG, blood pressure, oxygen saturation, end-tidal carbon dioxide (EtCO₂), and defibrillator pads if applicable to enable accurate continuous patient assessment throughout procedure.

5. Roles and responsibilities

Role	Responsibility
Airway MO/Duty Anaesthetist	Pre-Oxygenation using BVM/Mapleson Circuit and oxygen (O ₂) whilst preparation for intubation occurs Suction of oral secretions prior to intubation

Role	Responsibility
	Insertion of the endotracheal tube Confirm location of ETT by EtCO ₂ trace, observation of equal rise and fall of the chest, auscultation, SpO ₂ , chest x-ray (CXR)
Airway Assistant (Nurse/Anaesthetic Tech/MO)	Prepares intubation trolley with appropriate equipment Hands appropriate equipment to the Airway MO during the insertion of ETT Inflates the ETT Cuff once inserted Attaches Mapleson's Circuit to ETT or attached ventilator tubing to ETT Confirm location of ETT by EtCO ₂ trace, observation of equal rise and fall of the chest, auscultation, SpO ₂ , chest x-ray (CXR) Secures ETT
Assistant 3 (if applicable)	Applies cricoid pressure (if requested by MO)
Assistant 4 (Nurse/MO) (can also be primary nurse or Team Leader)	Drawing up of appropriate medications as prescribed by RMO Ensuring adequate Sodium Chloride 0.9% flushes are prepared Administration of appropriate medications as prescribed by RMO
Primary Nurse	Visually monitors patient observations to ensure hemodynamic stability Performs arterial blood gas (ABG) once intubation is completed to ensure adequate oxygenation and ventilation Ensures correct medications are prepared as prescribed by RMO
Scribe	Records vital signs throughout Document ETT size, position of ETT at teeth or lips, ventilator settings and cuff pressure and time of intubation. Documents volume, strength and quantity of medications administered during intubation procedure

5.1 Intubation

- The team leader is responsible for allocation of roles and for determining that the allocated staff understands the requirements of the role.

5.2 Post intubation procedure

- Perform an arterial blood gas (ABG) to assess patient's ventilation status (CO₂ and PaO₂ levels).
- Insert a nasogastric/oro-gastric tube PRIOR to Chest X-Ray.
- A portable chest x-ray is mandatory to confirm ETT placement.
- The ETT tip should be positioned approximately 2cm above the carina.

- Check ETT cuff pressure using the cuff manometer.
- Suction the subglottic port (if available) for secretions that may collect above the cuff.

5.3 Documentation

Document the events on the patient's ICU Flow chart:

5.4 Prescribe and sign for all medications administered on the medication chart as given throughout the patients' care.

Nursing Alert – Mandatory requirements

- All patients requiring mechanical ventilation must be cared for by a Registered Nurse who is deemed competent to manage an intubated patient.
- All nurses caring for patients requiring mechanical ventilation must have a nurse-to-patient ratio of 1:1.
- All mechanically ventilated patients will have continuous cardiac, oxygen saturation and end-tidal carbon dioxide (EtCO₂) monitoring with hourly documentation on the ICU flow chart.

6. Weaning patients from mechanical ventilation

Weaning is the process of decreasing ventilator support and allowing patients to assume a greater percentage of their own intrinsic ventilation.

A medical directive is required to wean the patient. A weaning plan for the patient will be discussed and decided on by the ICU multi-disciplinary team daily.

7. Nursing care and safety of the ventilated patient

- Check that the wall oxygen supply and suction equipment is available and functioning at the start of each shift.
- Ensure that prior to connecting a patient to the mechanical ventilator; the ventilator has passed a "device check".
- Check the re-breathing bag (Modified Mapleson's C bag) with valve system (**Figure 2.**) and / or self-inflating bag valve mask kit (**Figure 3.**) are assembled and in working order at head of bed. Ensure a facemask suitable for Bag Mask Ventilation (BMV) is in the bedspace – required in the case of accidental extubation.



Figure 2



Figure 3

- Check that emergency airway equipment is assembled
- Ensure the ventilator alarm settings are active and appropriate alarm limits for the patient have been set.
- Ensure that the patient's endotracheal tube is secured to the patient at the designated position, with woven adhesive tape, white cotton tape or Endotracheal Attachment Device (ETAD) as appropriate. Promote patient comfort by ensuring appropriate use of intravenous sedation and analgesia as prescribed.
- Prepare ongoing sedation/analgesia as prescribed ensuring adequate prepared medications are available for continuous administration.
- Assess patient pain post intubation, at the start of the shift and as required. Refer to AK-PRO-0339-17 Pain Management Procedure Policy using non-verbal pain assessment methods. Document findings as required in the ICU flow chart.
- Change the position of the endotracheal tube to the opposite side of the patient's mouth once per day or as required using aseptic technique (see section 6.3 - Repositioning ETT Tube – Side to Side (Two-person Procedure))
- Assess and document the area around the ETT securement device/tape/string for evidence of pressure injury daily.
- Following the repositioning record the new position and depth of ETT on the ICU Flow Chart.
- Stress / peptic ulcer prophylaxis as indicated – NOT ROUTINE PRACTICE.
- Seek a prescription for aperients if the patient does not have a bowel motion every third day
- Refer to Arterial Line: Management of Policy for care of invasive monitoring devices. Perform chest auscultation at least every two hours and record results on ICU flow chart. Perform Arterial Blood Gas (ABG) sampling as the patient's condition warrants and in the event of clinical deterioration.
- Clear secretions from patient's airway by suctioning as per the suctioning procedure. Ensure to regularly clear any build-up of secretions in the oral cavity. Record volume on the ICU flow chart after each suction.
- Clear secretions from the subglottic port (if available) of the ETT every two hours or as clinically indicated with a 10ml luer lock syringe. Record volume on the ICU flow chart after each suction. Ensure that cuff pressure is checked prior to each subglottic suction.
- All patients who are mechanically ventilated require DVT prophylaxis unless clinically contraindicated.

- All patients who are mechanically ventilated and sedated require daily sedation holds to facilitate assessment unless explicitly requested and documented by the treating Consultant.
- Document full neurological observations using the Glasgow Coma Scale (GCS). Refer to AK0176/24 Full Neurological Observations Guideline.
- All ventilated patients require hourly RASS assessment which must be documented on the ICU flow chart and four hourly minimum CAMS-ICU assessment.
- Commence and maintain enteral nutrition as prescribed by the MO or dietician. This may require measurement of the patient's ulnar length to estimate height. If the patient is intubated after hours, refer to AK0157/24 Enteral Feeding and Enteral Tubes Management – Adult procedure Policy appendix 3 Intensive Care Unit After-hours Enteral Feeding Protocol.
- If a patient has NGT/OGT insitu, measure and record the length of catheter each shift, comparing with previous measurement to ensure tube has not migrated. If in doubt, notify MO and do not use until the position has been re-confirmed.
- Assess area around NGT/OGT for evidence of pressure injury. Replace securement tape daily, ensuring that skin is prepared with an appropriate skin preparation wipe.
- All patients ventilated for two or more days, must have Physiotherapy review and the use of electrical muscle stimulation (E-Stim) must be considered.
- All patients ventilated for two or more days, must have the need for hand and feet splints explored, with referral to Physiotherapy and Occupational Therapy service as required.
- Maintain oral hygiene and oral mucosal hydration by providing mouth care at least two 2 hourly. Maintain an elevation of at least 30degrees of the patients head to prevent aspiration unless clinically contra-indicated (neurological or spinal injury).
- Assess the patient routinely for pressure injuries and ensure that the patient is regularly (standard three hourly) turned to alleviate pressure on at risk areas. Escalate concerns to the shift coordinator or MO as appropriate. Apply pressure relieving dressings and moisturising pH neutral emollient to at risk areas. Use pillows or pressure relieving devices to offload pressure on at risk areas.
- On admission and throughout, frequently assess and document the patient's full skin integrity and include preventative measures undertaken to avoid shear and pressure injuries.
- Maintain hygiene of the patient with a full wash during the night shift, and a 'freshen up' in the PM of the day. Include routine (daily minimum) maintenance of nails, hair, perianal area and indwelling catheter insertion site.
- Provide an explanation and give reassurance about mechanical ventilation to the patient, their relatives and /or significant others.

7.1 Documentation

- Document the patient's haemodynamic parameters and oxygen saturations hourly on the AKMR 174C ICU flow chart. Check the ventilator settings at the commencement of each shift and document settings on ICU Flow Chart
- Monitor and record all appropriate ventilator settings and patient parameters hourly / as required on ICU Flow Chart. Consider the need for additional ventilator measurements such as pPlat or totPEEP.
- Observe and record, the size and type of Endotracheal Tube (ETT)

- Observe and record the position of ETT at the patient's teeth/gums, noting which side of the mouth the ETT is positioned.
- These findings should be recorded on the ICU flow chart at the time of insertion, after any repositioning and at the beginning of each shift.
- Measure and record the endotracheal tube cuff pressure at the beginning of the shift, prior to subglottic suctioning, and as required on the ICU flow chart.
- Prevent complications such as aspiration, ischaemia, stenosis, inflammation or ulceration of the trachea by maintaining the endotracheal tube cuff pressure between 20 - 30cm H₂O, where possible.
- Monitor and document intravenous and enteral intake every hour.
- Document bowel motions every shift.
- Document hourly any fluid volume removed from the patient including from indwelling catheter, nasogastric drainage/aspirate, invasive surgical drains, or fluid removal due to CVVHDF.
- Calculate total input and output at the end of each 24hr period.
- Complete daily Braden Risk Assessment.
- Complete FRAMP daily.
- Ensure ICU Care plan is checked and maintained every shift.

Note: If the seal **cannot** be maintained at 20 - 30cm H₂O, increase the pressure to obtain seal and inform the MO.

7.2 Humidification

- Prevent thickening and retention of secretions and drying of the tracheal mucosa by ensuring the use of a water bath humidifier e.g. Fisher and Paykel® for all ventilated patients.
- Ensure that the heated humidifier is set to the ETT setting for all intubated patients.
- Ensure the level of sterile water in heated humidifier is maintained, changing the bag as necessary.
- Position and secure the ventilator tubing to minimise drag on endotracheal / tracheostomy tube.

7.3 Repositioning ETT Tube – Side to Side (Two-person Procedure)

Do not deflate cuff as this increases the risk of aspiration, air leak and ETT migration.

Ensure that either an ICU Consultant or fully intubation competent ICU Registrar is in the unit in the event of un-intentional extubation. Resuscitation and Airway trolley to be closely available and intubation kit prepared.

Repositioning of the ETT allows for assessment of skin and mucosal integrity which are prone to 'breakdown' due to tube movement and pressure from the tube.

- Person one holds the ETT securely in its current alignment ensuring that the tube stays at its current external measurement while Person two performs the next steps.

- Remove 'old tape' or string (if applicable – if Anchorfast securement device in situ this step is not necessary)
- Cleanse upper (maxillary), lower (mandibular) borders of the mouth opening and cheeks to remove excess oil and secretions from skin/old adhesive tape
- Move tube to opposite side of mouth. **Use tongue depressor to guide ETT**
- **Visually ensure** entire ETT length has moved to opposite side at back of mouth
- Secure ETT using appropriate tapes or securement device as outlined per procedure(s) section

8. Checking ETT cuff pressure

8.1 Introduction

Note: Patients admitted via Royal Flying Doctor Service (RFDS) may have the ETT cuff inflated with water. The water **MUST** be removed and replaced with air. This is a two person technique.

8.2 Equipment

- Yankauer sucker
- Y-suction catheter size 12F or 14F if unable to access oropharyngeal space with Yankauer
- High flow suction wall outlet on and in working order
- Cuff manometer **figure 4**
- Gloves (non-sterile)
- PPE as per standard precautions



Figure 4: Cuff manometer showing safe zone (solid green section) for ETT and tracheostomy cuff measurements

(NB: The solid section shows a range between 22-32cmH₂O, however adopted practice is 20-30cmH₂O being the accepted range).

8.3 Procedure

- Measure cuff pressure using cuff manometer at commencement of shift, prior to subglottic suctioning and/or when cuff leak suspected. If leak suspected one to two hourly measurement of cuff pressure is needed, until the issue is resolved or ETT is changed.
- Check the previous documented cuff pressure
- Don PPE as per standard and/or transmission-based precautions
- Suction oropharyngeal secretions prior to checking ETT cuff pressure

- Apply cuff manometer to the ETT pilot tube and measure the cuff pressure – inflate / deflate as required to ensure pressure in cuff pressure between 20-30cm H₂O as per the solid green section on the cuff manometer
- Document cuff pressure
- If excessive pressure or frequent re-inflation of the pilot balloon is required to prevent / manage a cuff leak, monitor Tidal Volumes, inform MO.

9. Securing the endotracheal tube

This is a two person procedure. One of the nurses must have experience securing ETTs.
Do not tape pilot balloon line to ETT. Ensure that the pilot line is not kinked or severed
Do not use alcohol wipes to clean ETT as it removes markings/numbers from ETT

The ETT must be secured to facilitate optimal ventilation and to avoid displacement or unplanned extubation.

Note: Prior to undertaking the procedure the nurse will ensure that all emergency equipment is available and in good working order and a MO competent in endotracheal intubation is available in the unit.

9.1 Equipment and personnel

- PPE as per standard and/or transmission-based precautions
- Gloves (non-sterile)
- Securing tapes – adhesive tape/foam tapes/white cotton tape/Anchorfast securement device. Scissors if required.
- Tongue depressor
- Facial cleansing equipment
- Shaving equipment (if appropriate)
- Skin preparation wipe
- Protective sheet
- Assistant = A second Registered Nurse)
- Cuff manometer
- 2 x oropharyngeal airway (size 3 and 4)
- Oxygen tubing and supply
- Re-breather bag (BVM)
- High flow suction wall outlet
- Yankauer sucker
- Y suction catheters (size 12F and 14F). Allows for management of the patient with an ETT in the event of tube dislodgement

9.2 Procedure

Tapes will be changed a minimum of once daily and ETT repositioned during daylight hours (as patient condition allows).

- In all areas the ETT will be secured

- Don PPE as per standard and/or transmission-based precautions Explain procedure to patient (if appropriate)
- Place patient on their back, Flat or sitting position
- Suction the ETT if indicated. Refer to [AKG Suction procedure](#)
- Suction oropharynx with Yankauer sucker to decrease risk of aspiration
 - Where ETT has above cuff suction port(yellow) this should also be aspirated
- Assess and check documented position of ETT in cm at teeth/gums
- Prepare all equipment
- Instruct assisting nurse to firmly hold ETT in position while tapes are applied/changed.
- Ensure documented level at teeth/gums is maintained throughout procedure

9.3 Cotton tape method (figure 5)

- Fold cotton (twill) tape provided in intubation set, into 1/3 to 2/3 ratio
- Loop folded end around ETT tube, feed ends through loop and tighten around ETT.
- Feed long end of tape under patient's neck and secure with knot followed by a bow on side of patient's face.
 - This allows for maximum contact between the tape and the ETT, therefore reducing ETT movement.
 - The tape should be tight enough to allow for two fingers to be placed between tape and patients face. This allows for the ETT to be easily unsecured if required at later time, and that the loose ends are not left dangling and therefore potential for getting caught up on other equipment



Figure 5 (A and B) cotton tape

9.4 Adhesive tape method

- Shave facial hair (if appropriate) – Verbal consent from patient or N.O.K should be obtained. Document consent on ICU flowchart and progress notes.
- Cleanse upper (maxillary) and lower (mandibular) borders of the mouth opening to remove excess oil and secretions from skin/old adhesive tape
- Apply skin preparation wipe around mouth opening (helps increase adhesion)
- Cut **two** strips of adhesive tape approximately 20cm long
- Cut each strip of tape in half along two thirds of the longitudinal length to form a Y-shape (see figure 6)
- Secure wide section of adhesive tape to cheek with split in tape (Y-shaped arm) sitting at the corner of the mouth nearest the ETT (figure 7)

- Attach one Y-shape arm of adhesive tape to the upper maxillary border and wrap the other Y-shape arm around the ETT (refer to figure 7)
- Attach second Y-shaped arm of adhesive tape to the lower mandibular border and wrap the other Y-shape arm around the ETT (refer to figure 8)

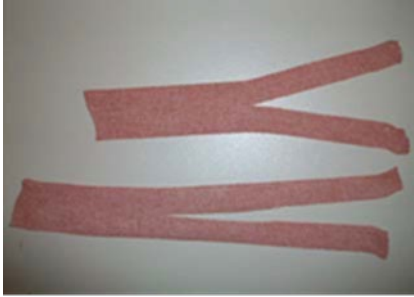


Figure 6: Adhesive tape



Figure 7 & 8 adhesive tape attached to upper maxillary border

9.5 AnchorFast method

9.5.1 Introduction

The AnchorFast tube fastener secures oral endotracheal tubes ranging in size from **5 to 10 mm** in diameter. The suitability of the oral endotracheal tube fastener must be assessed for each patient.

9.5.2 Application

Step 1 Prepare the skin— shave patient where necessary to ensure maximal adhesion

- Make sure the skin is clean, dry, and free of oily residue
- Do not use skin gel wipes or other brands of skin preps with the oral endotracheal tube fastener

Step 2 Remove the release liners

- Remove the release liners from the two skin barrier pads

Step 3 Place the device on the patient (figure 1)

- Centre the device on the upper lip so the non-absorbent upper lip stabilizer lightly touches the skin
- Position the “*one-click security clamp*”, approximately 1/2 inch below the patient’s upper lip
- Press the two skin barrier pads against the patient’s skin
- Hold the device in place until the barrier pads adhere well. This should take approximately thirty seconds



Figure 9⁷

Step 4 Apply the adjustable neck strap (Figure 10.7)

- Insert the narrow end of the strap through the plastic loop on the track
- Fasten the narrow end of the strap using the hook and loop closure
- Adjust straps on either side for added comfort and security
- Do not overtighten

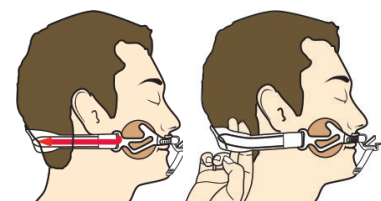


Figure 10⁷

Figure 11⁷

- Allow two fingers width between the strap and the back of the patient's head (Figure 11.7)

Step 5 Secure the endotracheal tube

- Squeeze the tabs on the sides of the gliding tube shuttle and move the clamp along the track to a location above the tube (Figure 12.7)
- Remove the release liner from the ET tube wrap exposing the adhesive. Before applying the wrap to the tube, make sure the tube is dry

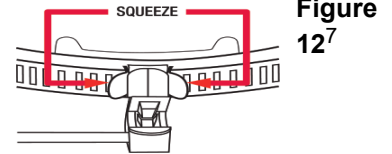


Figure 12⁷

and free of any residue

- Position the tube under the nonslip grippers
- Care should be taken to avoid aligning the inflation lumen directly under the nonslip grippers when securing the tube. Loop the wrap tightly around the tube, and pull the remaining portion of the wrap through the security clamp



Figure 13⁷

Secure the wrap by snapping the one-click security clamp shut (an audible click will be heard) (Figure 13.7)

- To reposition the tube, squeeze the shuttle tabs on the outer edges and move in either direction along the tube track (Figure 14.7)
- Reposition the tube, side to side, at least once per shift or more frequently if the patient's condition dictates, to minimize the risk of injury to the skin and/or lips from unrelieved pressure
- After repositioning, ensure ETT has also been moved at the back of the oropharynx. If manual repositioning is required at the back of the oropharynx, care should be taken to prevent injury to the oral mucosa.

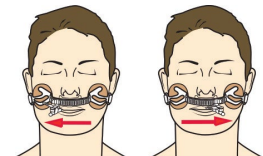


Figure 14⁷

9.6 Precautions

- Patients without front upper teeth or unable to wear upper dentures may lack the maxillary support required to use the oral endotracheal tube fastener
- Use caution in patients with full or swollen lips, facial swelling, dental appliances, restorative implants, and/or loose or protruding teeth
- Patients with facial hair may lack the necessary support to anchor the skin barrier pads so consider cotton tape.
- After application of the oral endotracheal tube fastener, check to ensure that both the oral endotracheal tube fastener and the endotracheal tube are secure and correctly positioned
- To ensure proper fixation of the device, exercise caution with the use of other devices and/or instruments (i.e., feeding tubes, fibre optic scopes within the oral cavity during endotracheal intubation)
- Reconfirm position, depth of intubation, and patency of the tracheal tube or other airway device during and after any change in the patient's head, neck, or body position, or any change in the location of the fixation device
- To minimize the risk of pressure injury, inspect the patient's lips and skin at least every two hours or more frequently if the patient's condition dictates

- Be sure to frequently assess device fixation/adhesion since wear time varies by patient
- Discontinue use of the device if redness or skin irritation occurs
- Repeated adjustment of the endotracheal tube in a distal or proximal direction may affect the performance of the ET tube wrap
- Care should be taken to avoid aligning the inflation lumen directly under the non-slip grippers when securing the tube
- After use, dispose of gloves and device in the Clinical Waste bin.

9.7 Post-securement of ETT

- Ensure ventilator tubing is supported to minimise tube migration / dislodgement. Use support arm attached to ventilator or the Perspex tube holder.
- Position ventilator tubing to collect condensation away from the ETT – i.e. not to drain into patient
 - Auscultate chest and record on ICU Flow Chart two hourly standard.
 - Document change of ETT tapes and position and level of ETT in cm at teeth/gums on ICU Flow Chart and Patient Care Plan

10. Use of restraints for mechanically ventilated patient in ICU

Routine weaning from mechanical ventilation **does not** require every patient to be restrained as it is natural that the patient will find the endotracheal tube (ETT) uncomfortable and difficult to tolerate. The patient may reach out and attempt to remove the endotracheal tube, this is at great risk to the patient's airway and ability to breathe.

Some patients are delirious when first waking and may pull at critical invasive lines (arterial line, central line with inotropes to support blood pressure etc.), which is a risk to their cardiovascular system.

This cohort of patient can be managed with reassurance and keeping them informed of the plan of care.

In Western Australia the circumstances under which a patient may be restrained include

- For urgent assessment and/or treatment of patients (excluding psychiatric treatment of mental illness) who are unable to make reasonable decisions for themselves and are at imminent risk of causing serious harm to self or others
- In situations where the patients has been assessed to not have capacity, to prevent immediate risk of harm to self or others (Doctrine of Necessity). See [Use of Restrictive Practices in Non-Authorised Healthcare Settings Policy](#)

Restraint should never be used for the convenience of staff or to overcome a lack of adequate supervision. Nor should restraint ever be used to punish or negatively reinforce problem behaviours. The first line of management when de-sedating an intubated patient, is constant reassurance, explanation of the purpose of the invasive line, endotracheal tube, reorientation and keeping the patient informed of their plan of care.

10.1 Considerations

- The least restrictive form of restraint should be used and only as a temporary solution.

- Restraint may only legitimately be used to assess/address the imminent risk a person poses to him/herself, to others or to property. The initiation of restrictive practices is often required at short notice to address an imminent or actual risk and a potential emergency situation. The use of restrictive practices, particularly physical and/or mechanical restraints, must be proportionate to the imminent risk of harm, applied safely using the minimum amount of force considered necessary and reasonable, for the minimum duration necessary.
- Restrictive practices should only be considered for patients that have been identified as a risk to themselves and others e.g. at risk of self-extubation or pulling / removing infusion lines, previous episode of violence and aggression, patients with severe psychiatric disorders, alcohol and drug withdrawal and/or delirium. These patients should receive treatment for the underlying condition prior to de-sedation and weaning occurring.

- Restrictive practices should be used for the shortest period of time possible, and in conjunction with constant reassurance, explanation of the purpose of the infusion lines and endotracheal tube, reorientation and keeping the patient informed of their plan of care.
- Restraints should be applied as per relevant AKG Policy

10.2 Nursing Management

Once a restrictive practice is applied, the clinical team must ensure ongoing patient assessments are conducted, as required, in accordance with their care requirements and according to AKG policy.

The use of restrictive practices must be reassessed immediately if there is an injury or deterioration of the patient's health condition(s).

HSPs must undertake a review of the patient's deterioration to ensure that any continuation in the use of restrictive practices considers the level of harm, injury or ongoing risk to the patient or others.

Ensure the plan to wean is documented in patient integrated notes by the MO.

Unplanned extubation of patients who require mechanical ventilation can be life-threatening; the most serious consequences include cardiopulmonary arrest and death.

The Nurse is to consider the following when assessing if the patient is a risk for self extubation:

- Consider the reason for initial intubation
- History – behavioural (previous attempts at de-sedation and extubation)
- Comorbidities pre intubation such as psychosis or head injury.
- History of drug and/or alcohol use
- Restrained patients can still extubate themselves

An ICU Nurse may initiate restraint in an emergency if assessment indicates the necessity. Refer to above section 9 and 9.1.

Once restraints are initiated, it must be authorised and documented by the MO.

10.3 Documentation and reporting

Where there is an ongoing need for the use of a restraining device, a care management plan must be developed by the clinical team documenting the requirement for the practice, the patient's decision-making capacity, behavioural management and de-escalation strategies utilised. The plan can be amended at any time following a review of the patient's care by the clinical team

Each use of a restrictive practice must be documented in the patient's medical record and restraint monitoring tool. This must include the following information:

- Details of the situation, including the patient's mental state, deterioration,
- associated behaviours and the level and type(s) of risk
- Unsuccessful lower risk de-escalation strategies utilised and reasons for failure, if applicable
- The type of restrictive practice used and clinical rationale
- Assessment of mental capacity and the outcome of that assessment
- The use of supplementary support from security officers, if applicable.
- Any physical injuries resulting from the use of a restrictive practice to patient or clinician.
- Notification of family member, guardian, enduring guardian or substitute decision-maker, if appropriate.
- Any medications administered
- Any multidisciplinary reviews conducted

Form AKMR147E - Physical Restraint Flowsheet must be signed by the MO to authorise the use of restraints and to document patient safety observations.

11. Extubation

Extubation refers to the removal of the endotracheal tube and should not be considered until it has been determined that the patient is able to protect their own airway and the airway is patent or a decision to palliate has been confirmed.

The decision to extubate a patient is the responsibility of the Intensive Care Consultant.

Extubation may be a discontinuation of mechanical ventilation as the patient is considered to be able to manage their own ventilation without an ETT. Generally, the initial reason for the patient requiring intubation no longer exists.

Extubation may also be a terminal event.

Post-extubation, patients should initially be kept Nil by mouth with medical direction given on when to allow oral intake. Under most circumstances, this will be four hours post-extubation.

All patients should be placed on oxygen after extubation, with the mode of delivery to be determined before extubation proceeds.

Purpose (Planned Extubation)

To remove the ETT with minimal risk and discomfort and then allow the patient to ventilate independently. This requires an assessment of whether the patient can protect their airway (i.e. has a sufficient cough and adequate level of consciousness), the volume of respiratory secretions, and whether the airway is patent (i.e. a cuff leak is present) and respiratory effort.

Ideally, patients are extubated awake, alert, following commands, and preferably off sedatives. However, this is not always feasible and a small proportion of patients who are cooperative and rousable upon stimulation or receiving minimal sedation may be extubated with vigilant monitoring of the airway and mental status and prompt weaning of sedatives.

11.1 Equipment and Personnel

- Intubation trolley set as per ICU guidelines.
- Y-suction catheters (size 12F and 14F) and Yankauer sucker
- High flow suction wall outlet on and in working order
- Manual re-breather bag with oxygen (O₂) tubing attached and running @ 10 litres per minute.
 - Ensure re-breather bag has been fully inflated. Enables emergency re-intubation if patient does not maintain spontaneous ventilation.
- Hudson mask with O₂ tubing attached and running @ 10 litres per minute
- 10mL syringe
- Scissors (to cut cotton tape if applicable)
- Gloves (non-sterile)
- PPE as per standard and/or transmission-based precautions
- Protective sheet (Bluey)
- ICU Senior Registrar or Consultant present on the unit during and after extubation

Personnel skilled in endotracheal intubation and insertion of other invasive airways should be immediately available whenever extubation is performed

11.2 Criteria for Extubation

A patient can be considered for extubation in general, but not limited to the following criteria:

11.2.1 Central nervous system:

- Richmond Agitation Sedation Score (RASS) -2 to +2
- Patient is cooperative
- Patient able to demonstrate adequate cough on demand and on suction
- Patient able to breathe deeply on request
- IV analgesia may continue, and sedation may be weaned and/or discontinued

11.2.2 Respiratory (Airway Protection)

- PaO₂ ≥ 70 mmHg on FiO₂ ≤ 0.4 with PEEP ≤ 5 cmH₂O and PS of ≤5cmH₂O
- PaCO₂ < 51 mmHg

- $\text{SaO}_2 \geq 95\%$
- $\text{pH} > 7.25$
- Respiratory rate < 24 per minute
- Tidal volumes $> 5\text{mL/kg}$
- Air entry equal (unless pre-existing pathology) on auscultation with minimal use of accessory muscles
- Easily managed secretions
- Chest x-ray reviewed by MO

The decision to extubate a patient is the responsibility of Senior Critical Care Medical Staff.

11.2.3 Cardiovascular – (Haemodynamically stable)

- Heart rate (HR) < 120 bpm
- Systolic blood pressure (SBP) > 90 mmHg
- Diastolic blood pressure (DBP) < 100 mmHg
- Warm and well perfused
- Not requiring high or rising inotropic support

11.2.4 Gastrointestinal

- Patient ideally nil per mouth/naso/orogastric tube for four (4) hours prior to extubation
- Gastric contents must be aspirated via naso/orogastric tube just prior to procedure to minimise the risk of aspiration.

11.2.5 General

- Patient unlikely to return to theatre in the short term
- Patient unlikely to undergo diagnostic procedures such as bronchoscopy, CT scan or transoesophageal echo in the short term
- Body temperature $> 36^\circ\text{C}$

11.2.6 Upper Airway Oedema

- Patients with suspected upper airway oedema may require the ICU Senior Registrar or Consultant to perform a cuff leak test
- Patients with suspected upper airway oedema may also require the ICU Senior Registrar or Consultant to perform a direct laryngoscopy prior to extubation

11.3 Extubation – Procedure

- Check that the patient's respiratory and cardiovascular parameters, and arterial blood gas (ABG) results meet the respiratory and cardiovascular criteria
- Consult with ICU/ ED Senior Registrar or Consultant regarding intention to extubate
- Inform shift coordinator of impending extubation
- Place intubation trolley within close proximity and gather necessary equipment
- Check the anaesthetic chart/ICU charts for laryngoscopic grade of intubation.

If Grade III or IV alert ICU Senior Registrar or Consultant and ensure difficult intubation trolley is placed within proximity. Laryngoscopic grades of intubation as per Anaesthetic Chart/Patient integrated notes

- Don PPE as per standard and/or transmission-based precautions
- Ensure patient has continuous ECG and SaO₂ monitoring
- Explain procedure to patient/family/carer to alleviate anxiety and ensure the patient understands their role in the procedure (as appropriate)
- Aspirate naso/orogastric tube to minimise the risk of aspiration
- Pre-oxygenate patient with FiO₂ of 1.0 (100%) for 3 minutes prior to extubation
- Allow patient to settle after the coughing induced by the suctioning (approximately 2 minutes on FiO₂ of 1.0)
- Place patient in an intermediate or high Fowler's position unless clinically contraindicated (as per Patient Care Plan)

Patients being extubated for palliative reasons by medical staff will not necessarily meet or should have to meet the extubation criteria. Refer to [End of Life Care Procedure Policy](#)

- Suction patient's posterior oropharynx with the Yankuer sucker to remove pooled secretions prior to cuff deflation
- Attach a 10mL syringe to the balloon of the pilot line
- Loosen the ETT securement device
- Instruct patient to inhale and deflate the cuff fully at peak inspiration
- Remove the ETT as patient begins to exhale/cough
- Do not extubate with the suction catheter in the ETT as this stimulates the cough reflex and may cause trauma to the vocal cords as patient coughs on the ETT
- Suction patient's oropharynx to remove excess secretions
- Have patient cough/talk. Intubation and extubation may result in vocal cord injury
- Place patient on a Hudson mask at 10 litres per minute, or as prescribed on Oxygen Prescription Chart. Ensure adequate oxygenation via face mask post-extubation
- Increase oxygen delivery modality should initial oxygen therapy prove inadequate
- Remove PPE and perform hand hygiene
- Obtain an ABG after 30 minutes and titrate FiO₂ accordingly

11.4 Post-Extubation

In ICU document date and time of extubation on the ICU Flowchart and on the Intubation/Extubation section on ICU Flow Chart and the patient integrated notes

- In ED document date and time of extubation and plan for patient on Triage Form
- Monitor and document on the ICU Flowchart/ED triage form:
 - Respiratory rate and pattern - presence of stridor
 - Air entry, SaO₂ and ABG
 - Heart rate and blood pressure
 - Continuous ECG monitoring

- Patients who are palliative may not require standard monitoring post-extubation.
- In ICU ensure catheter mount on re-breather bag has been replaced with the facial mask

Notify medical staff on presentation of any of the following:

- Signs and symptoms of airway obstruction
- Stridor
- Marked use of accessory muscles during respiration
- Tachycardia
- Deterioration in ABGs ($\uparrow$ PaCO₂, $\downarrow$ pH, $\downarrow$ PaO₂, $\downarrow$ SaO₂ assess these values in consultation with medical staff)
- Agitation
- Tachypnoea
- Inability to cough/swallow
- Increased secretions
- Laryngeal spasm/oedema or respiratory distress may necessitate re-intubation
- Patient to fast from oral intake for two to four hours post-extubation (MO to confirm)
- Patients who have been intubated for >6 days, nil orally until physical protective mechanisms have been formally assessed by MO

11.5 Complications/Potential Problems Following Extubation

11.5.1 Respiratory

- Hypoxemia after extubation may result from but is not limited to:
- Failure to deliver adequate inspired oxygen fraction through the natural upper airway
- Acute upper airway obstruction secondary to laryngospasm
- Development of post-obstruction pulmonary oedema
- Bronchospasm
- Development of atelectasis, or lung collapse
- Pulmonary aspiration
- Hypoventilation
- Hypercapnia after extubation may be caused by but is not limited to:
- Upper airway obstruction resulting from oedema of the trachea, vocal cords, or larynx
- Respiratory muscle weakness
- Excessive work of breathing
- Bronchospasm
- Airway protective reflexes may be depressed immediately following and for some time after extubation.

11.5.2 Cardiovascular

- Hypertension
- Hypotension
- Tachycardia

- Arrhythmias
- Myocardial ischaemia

11.5.3 Potential Long-Term Problems Following Extubation

- Laryngeal stenosis
- Granuloma/palsy
- Tracheomalacia
- Tracheal stenosis at site of the cuff or tube tip
- Upper airway obstruction from laryngospasm, laryngeal oedema or supraglottic obstruction
- Tracheo-oesophageal fistula

12. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- HDWA [High Risk Medication Policy MP 0131/20](#)
- HDWA [Clinical Incident Management Policy MP 0122/19](#)
- HDWA [Use of Restrictive Practices in Non Authorised Healthcare Settings Policy MP 0186/24](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS: 21](#)

13. Supporting information

The following documents support the implementation of this policy:

- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-22](#)
- [Suctioning Management Guideline AKG-GUI-430-11](#)
- [Clinical Incident Management Procedure AKG-PRO-0087-17](#) Royal Perth Hospital (EMHS) [Intubated Patient Management in the Intensive Care Unit \(ICU\) and Emergency Department \(ED\) Nursing Practice Standard](#)
- [Medication Management Procedure AKG-PRO-0255-19](#)

14. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

15. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.15 High-risk Medicines

- The health service organisation:
 - a. Identifies high-risk medicines used within the organisation
 - b. Has a system to store, prescribe, dispense and administer high-risk medicines safely
- Standard 5 – Action 5.35 Minimising restrictive practices restraint
- Where restraint is clinically necessary to prevent harm, the health service organisation has systems that:
 - a. Minimise and, where possible, eliminate the use of restraint
 - b. Govern the use of restraint in accordance with legislation
 - c. Report use of restraint to the governing body
- Standard 8 – Action 8.03 Partnering with consumers
- Clinicians use organisational processes from the Partnering with Consumers Standard when recognising and responding to acute deterioration to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making
- Standard 8 – Action 8.10 Responding to deterioration
- The health service organisation has processes that support timely response by clinicians with the skills required to manage episodes of acute deterioration

16. References

1. Branson, R. D. (2007) Secretion Management in the Mechanically Ventilated Patient. *Respiratory Care*, 52(10):1328-1347.
2. Disma, N., & Breschan, C. (2023). Perioperative monitoring: methods, implementation, and interpretation [image]. In Lerman, J. (eds) *Neonatal Anesthesia*. Springer, Cham. Retrieved from https://doi.org/10.1007/978-3-031-25358-4_7
3. Joanna Briggs Institute for Evidenced Based Nursing and Midwifery (2007) Artificial Airway: Suctioning. *Acute Care Practice Manual*.
4. Joanna Briggs Institute for Evidenced Based Nursing and Midwifery (2007). Endotracheal Tube (ventilated patient) Care. *Acute Care Practice Manual*.
5. Joanna Briggs Institute for Evidenced Based Nursing and Midwifery (2007) Tracheostomy: Routine Care. *Acute Care Practice Manual*.
6. Orebaugh, S., Snyder, J.V. (2024). Direct laryngoscopy and endotracheal intubation in adults [image]. Retrieved from <https://www.uptodate.com/>
7. Hollister Australia (2014). AnchorFast Guard [image]. Retrieved from <https://www.dansac.com.au/-/media/files/pdfs-for-download/australia/hl814-anchorfastguardcaretips.ashx>
8. Solomita, M. and Smaldone, G.C. (2007) Humidification and Non-invasive Ventilation. *Respiratory Care*, 52(1):24-25.
9. St. John, R.E. (2004) Airway Management. *Critical Care Nurse*, 24(2):93- 96.
10. Student Nurse Life (2022). Critical Care Setting Haemodynamic and Respiratory Monitoring: Capnography in Critical Care. [image]. Retrieved from <https://student-nurse-life.com/critical-care-setting-haemodynamic-and-respiratory-monitoring/>

17. Glossary

The following definitions are relevant to this policy.

Term	Definition
Mechanical ventilation	Mechanical ventilation is the medical term for artificial ventilation where mechanical means is used to assist or replace spontaneous breathing
Endotracheal intubation (intubation)	Intubation is the process of inserting a tube through the mouth or nose into the trachea to establish a patent airway.
Extubation	The process of removing an endotracheal tube
Doctrine of necessity	A common law doctrine that allows an individual to perform certain actions that would normally be deemed as contraventions of the law. The doctrine is only applicable in cases of imminent risk of serious harm or death to self or others
Restraint	The restriction of an individual's freedom of movement by physical or mechanical means.

18. Document contact

Enquiries relating to this guideline may be directed to:

Title: Nurse Unit Manager

Department: Intensive Care Unit

19. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

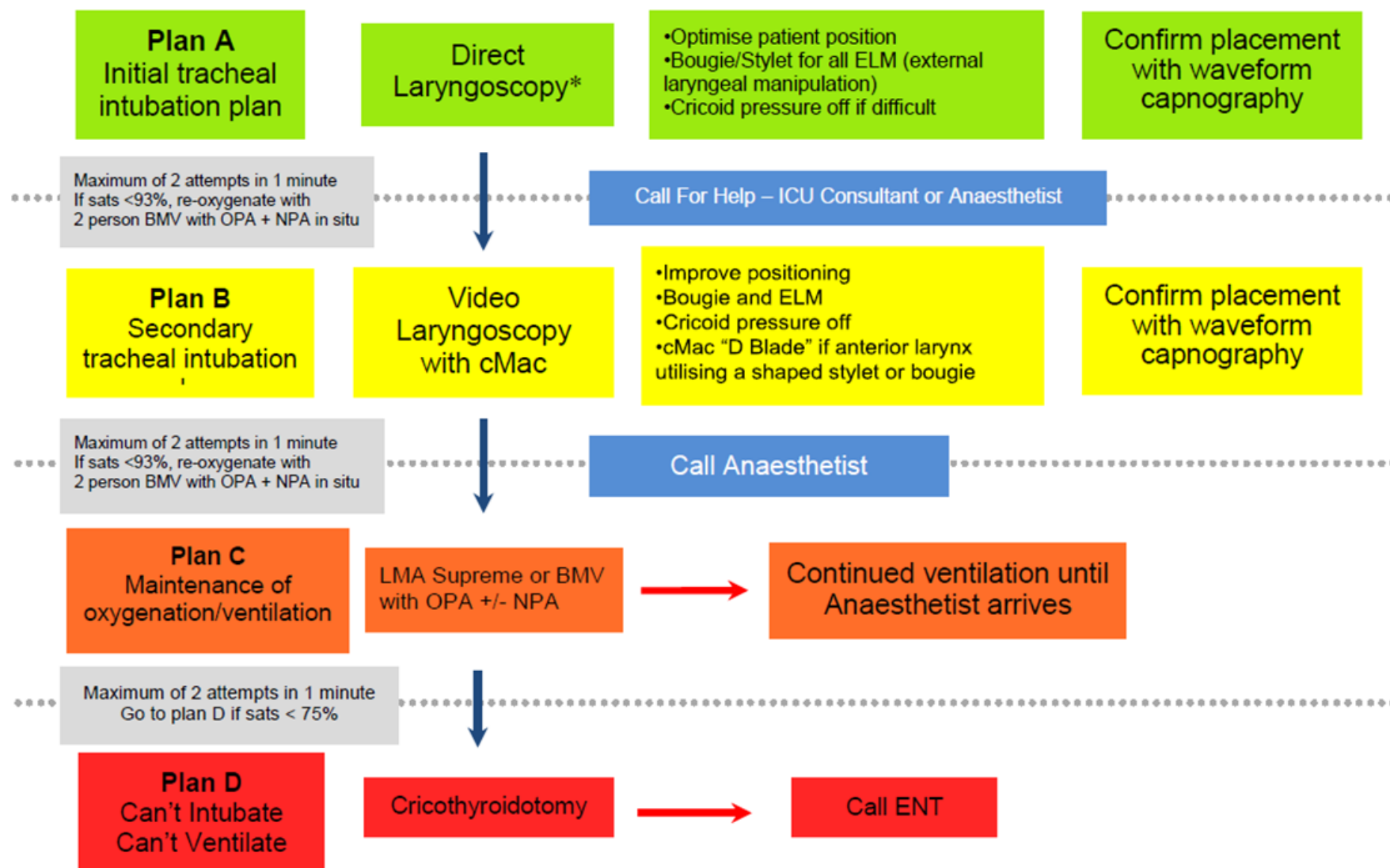
Version	Effective from	Effective to	Amendment(s)
V3	02/04/2025	02/04/2028	Scheduled review

20. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – Director Clinical Services
Authorisation date	01/04/2025
Published date	01/04/2025

Appendix 1: ICU Intubation algorithm



Appendix 2: Capnography

Observe EtCO₂ trace on monitor and ensure it resembles figure 11.

The capnogram provides

- A visual verification of proper ETT placement
- Validation of EtCO₂ value
- Assessment of the patient's ventilator/breathing circuit integrity
- **Phase I:** (Inspiratory baseline) reflects inspired gas, which is normally devoid of CO₂ (shown as A to B on the capnogram)
- **Phase II:** (Expiratory upstroke) is characterised by a rapid rise in CO₂ concentration as anatomical dead space is replaced with alveolar gas, leading to Phase III (B to C on the capnogram)
- **Phase III:** Is the alveolar plateau (C-D). The EtCO₂ value displayed on the monitor is considered the value taken at end exhalation (point D on the capnogram)
- **Phase 0:** Is the inspiratory down stroke, the beginning of the next inspiration (D to E). (Refer to figures 9 and 10).

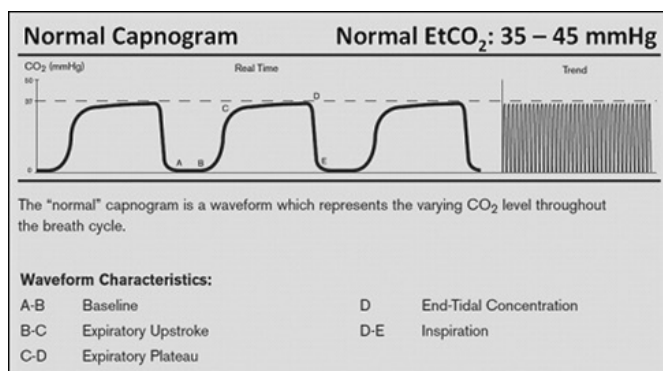


Figure 15²: Normal Capnogram

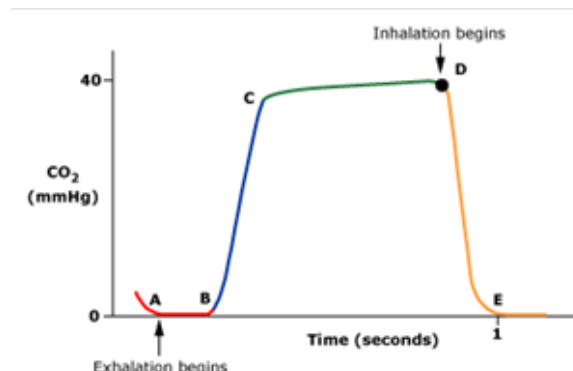


Figure 16¹⁰: Normal CO₂ Waveform

Abnormalities in the monitor trace can be indicative of incorrect ETT placement e.g. inadequate cuff seal, oesophageal intubation, airway obstruction, bronchospasm, inadequate ventilation settings/malfunction.

NB. If ETT insertion is unsuccessful the patient is re-oxygenated with BVM before attempting another insertion.

Prolonged attempts at airway insertion can result in reduced oxygenation, which can then further compromise the patient clinically.

Appendix 3: Richmond agitation sedation scale (RASS)

Term	Description
Combative	Overtly combative/violent. Immediate danger to staff e.g. punching, thrashing around bed
Very Agitated	Pulls on or removes tube/s, catheter/s or has aggressive behaviour towards staff e.g. purposeful or localising
Agitated	Frequent non-purposeful movement or patient - ventilator dysynchrony E.g. non-localising
Restless	Anxious or apprehensive but movements not aggressive or vigorous e.g. plucking, chewing ETT
Alert & Calm	
Drowsy	Not fully alert but has sustained (>10seconds) awakening, with eye contact, to voice
Light sedation	Briefly (<10seconds) awakens with eye contact to voice
Moderate sedation	Any movement (but no eye contact) to voice
Deep sedation	No response to voice, but any movement to physical stimulus E.g. Coughs with turning, bites or sucks with mouth care
Unroutable	No response to voice or physical stimulation

Process

- Observe the patient.
- Is patient alert and calm? (Score 0)
- Does patient have behaviour consistent with restlessness or agitation? (Score +1 to +4 using criteria listed above)
- If patient is not alert, in a loud speaking voice state patient's name and direct patient to open eyes and look at speaker. Repeat once if necessary.
- Can prompt patient to continue to look at speaker.
- Patient has eye opening and eye contact that is sustained for > 10 seconds (Score –1)
- Eye opening and eye contact, but this is not sustained for 10 seconds (Score –2) Patient has any movement in response to voice; excluding eye contact (Score –3)
- If patient does not respond to voice, physically stimulate patient by shaking shoulder and/or rubbing sternum.
- Patient has any movement to physical stimulation (Score –4). Patient has no response to voice or physical stimulation (Score –5)